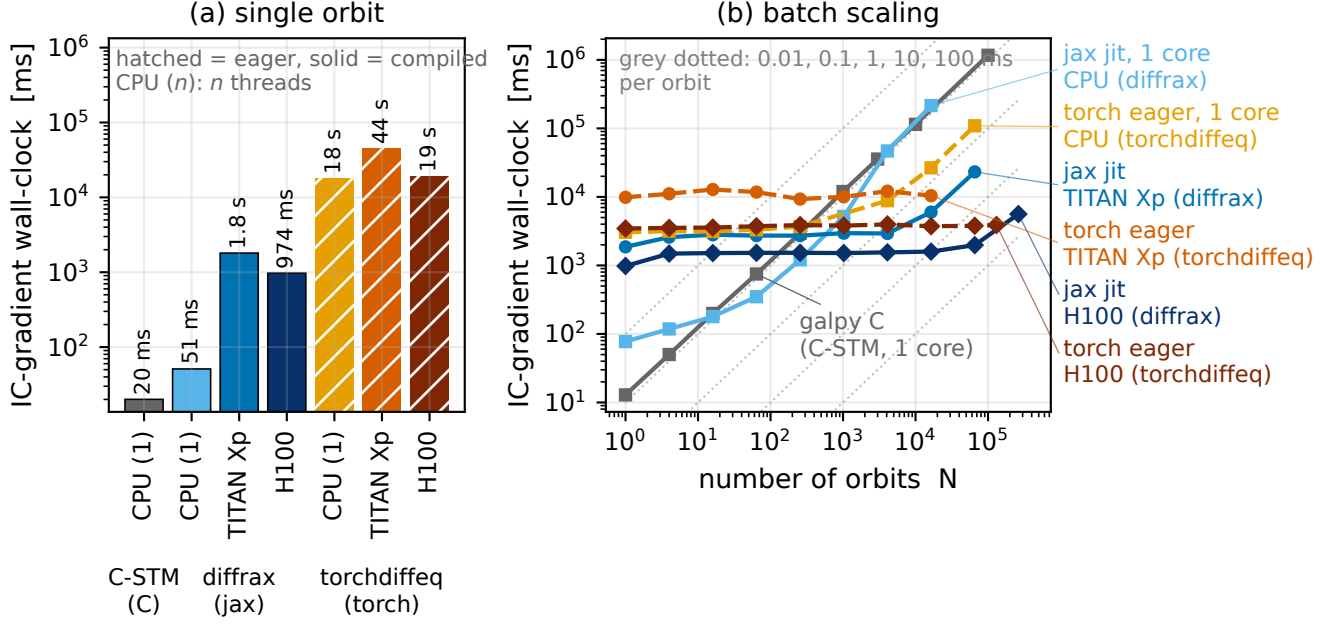
(a) single orbit
hatched = eager, solid = compiled
CPU (n): n threads
IC-gradient wall-clock [ms]
20 ms
51 ms
1.8 s
974 ms
18 s
44 s
19 s
CPU (1)
CPU (1)
TITAN Xp
H100
CPU (1)
TITAN Xp
H100
C-STM (C)
diffrax (jax)
torchdiffeq (torch)
(b) batch scaling
grey dotted: 0.01, 0.1, 1, 10, 100 ms per orbit
galpy C (C-STM, 1 core)
number of orbits N
jax jit, 1 core CPU (diffrax)
torch eager, 1 core CPU (torchdiffeq)
jax jit TITAN Xp (diffrax)
torch eager TITAN Xp (torchdiffeq)
jax jit H100 (diffrax)
torch eager H100 (torchdiffeq)